



Simulation of avascular tumor growth and drug response in a microfluidic device with a cellular automaton model

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Abstract

The microfluidic system is capable of recapitulating key attributes of *in vivo* circumstances and, therefore, becomes a valuable platform for better understanding tumor growth dynamics and evaluating drug efficiency. While numerical simulations have been envisioned as powerful tools for validating versatile performance of advanced microfluidic platforms, cell growth within these microchannels has not yet been theoretically modeled. In this paper, we developed an experimental data-driven cellular automaton model, which was adopted for simulating cell behaviors and drug responses in a microfluidic system. The boundaries of the cellular automata lattices and prohibited zones for simulation were directly converted from microscopic images of cell morphology and the microchamber configuration. The dynamic progression of tumor growth at the avascular stage was predicted by incorporating the biophysical and molecular characteristics of cells and their interactions with surrounding environment. The simulated proliferation rate of tumor cells over time demonstrated its dependency on nutrient delivery, aligning well with experimental observations in the microfluidic culture. The spatiotemporal efficacy of the chemotherapeutic compound doxorubicin (DOX) on the microfluidic culture was also simulated. The similarity between *in silico* simulations and *in vitro* tumor response upon drug interaction highlighted the potential of the computational models as complementary tools for predicting the drug treatment efficacy with acceptable accuracy before practical applications.

Keywords Microfluidic culture · Numerical simulations · Tumor growth · Cellular automaton model · Drug evaluation

1 Introduction

Tumorigenesis is a complex and multifactorial process that involves various genetic alterations or epigenetic mutations. During the initial stage of tumor development, tumor cells rely on the diffusion of nutrients and oxygen from surrounding stroma for their survival and growth. The primary tumor continues to grow following Gompertz-like curves until it reaches a saturated volume of 1–2 mm³ (Szabó and Merks 2013). As the tumor grows in size, tumor cells would stimulate the formation of new vascular networks through

angiogenesis to meet the nutrient demand and transit into the vascular growth phase (Tang et al. 2014). Understanding the behavior of tumor cells during the avascular phase is crucial for identifying characteristic signals that govern malignant tumor-related angiogenesis and metastasis. Sophisticated *in vitro* and *in silico* models have been developed to study the avascular growth of tumors, offering rigorous and reliable approaches to investigate the spatiotemporal aspects of tumorigenesis that may be challenging to discern from *in vivo* experiments (Milotti and Chignola 2010; Taghibakhshi et al. 2019).

Compared to conventional monolayer cultures, tumor cells grown under three-dimensional (3D) conditions can better preserve their physiological phenotypes and exhibit physiologically relevant functions through intercellular communications and interactions with their extracellular microenvironment. As such, multicellular 3D tumor aggregates, in particular, have demonstrated immense potential as important *in vitro* models for studying the pre-vascular phase of solid tumors for screening therapeutic agents and developing new anticancer strategies

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(Han et al. 2021). Recent advances in microfluidic techniques own the ease of formulating a dynamic yet controllable microenvironment, facilitating long-term cell growth. Microfluidic-based 3D tumor tissue models have successfully demonstrated their utility for cancer therapeutic testing (Uhl and Liu 2019). In our previous work, we reconstructed reproducible 3D tumor tissue structures with layering architectures within a microfluidic device by continuously supplying fresh culture media and confining cell growth within microchambers. Furthermore, by integrating a fluorescence detection and imaging system into the microfluidic platform, we were able to directly monitor tumor progression and drug response in a real-time fashion (Yang et al. 2015, 2020).

Alongside the progress in experimental technologies for in vitro 3D tumor models, computational tools have also played an important role as complementing approaches in predicting cell growth and interaction in relation to cancer development. Theoretical computations can help determine physical phenomena and critical parameters that are challenging to be measured using traditional experimental methods (Chen et al. 2015). Numerical simulations have also aided in evaluating the effectiveness of microfluidic systems in facilitating dynamic cell growth and optimizing microchannel designs at a lower cost (Carvalho et al. 2021). Previously, we used finite-element modeling (FEM) to simulate velocity distributions within different microfluidic geometries and analyzed the influences of structural parameters of vasculature microchannels on blood flow dynamics using the computational fluid dynamics (CFD) method (Yang et al. 2021; Dong et al. 2022). However, the behaviors of cells within these microchambers remained to be theoretically modeled through computational approaches.

The cellular automaton model (CAM) is a mathematical framework that focuses on the complex behavior of individual cells, as well as their intricate interactions with neighboring cells and the surrounding microenvironment. This model consists of a grid of cells, and the states of these cells are determined by the states of their neighboring cells, according to a set of given rules (Swat et al. 2012; Balasteros Hernando et al. 2019). By applying these rules at each time step, CAM can simulate the dynamic behavior of cells in the system and explore the emergent properties that arise from the interactions between the cells. Nivlouei et al. revealed the capability of a cell-based numerical simulation of tumor progression to explore the outcomes of tumor-targeted therapy by incorporating intracellular, cellular, and extracellular scales (Jafari Nivlouei et al. 2022). Brüningk et al. presented a 3D CAM to simulate the simultaneous radiation and hyperthermia treatments of the human colorectal carcinoma cell line HCT116, which was successfully calibrated and validated against experimental data (Sadhukhan and Basu 2020).

The cellular Potts model (CPM), also known as the Glazier–Graner–Hogeweg (GGH) model, is a frequently used CAM for investigating the growth and development of biological systems, particularly avascular tumors (Graner and Glazier 1992; Andasari et al. 2012). It incorporates the biophysical and molecular characteristics of tumor cells to provide a comprehensive understanding in portraying tumor cell proliferation, migration, and differentiation at different stages (Szabó and Merks 2013). Sheraton et al. examined the response of human cervical cancer HeLa-C3 cell spheroids to the combinatorial administration of two chemotherapeutic drugs using a CPM-based mechanistic mathematical model, highlighting the spatial mass-transfer kinetics and cell growth dynamics (Sheraton et al. 2020; Kaura et al. 2021).

In the present study, we incorporated true experimental data from biological experiments and microscopic observations into computational simulations to model the growth of avascular tumors over time and their response to chemotherapeutic drugs in a microfluidic chamber, using the CPM approach. The model, of which key parameters were directly derived from experimental results, not only reproduced the experimental data in the microfluidic system with spatiotemporal knowledge but also provided novel insights into tumor development and drug response that otherwise cannot be easily observed or measured with experimental techniques.

2 Materials and methods

2.1 Experimental model

2.1.1 Microfluidic device design, fabrication, and system setup

The design and fabrication process of the microfluidic device with eight parallel cell culture microchambers have been described in detail in our previous studies (Yang et al. 2015, 2020). The microfluidic device was mounted on a dual-modal optical probe spectroscopic system, enabling on-site spectra acquisition for therapeutic evaluation (Yang et al. 2022). The inlets of eight microchambers were connected to the culture-medium-containing syringes with polytetrafluoroethylene (PTFE) tubes and blunt-end needles. A digital syringe pump (NE 1000 New Era Pump Systems Inc.) was used to control the flow rates.

2.1.2 Dynamic cell culture in the microfluidic system

Human breast adenocarcinoma cell line (MCF-7) and human adipose-derived stromal cells (ASC) from American Type Culture Collection (ATCC, Manassas, VA) were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 100 units/mL penicillin,

and 100 mg/mL streptomycin. A mixture of MCF-7 and ASC cell suspensions at a ratio of 1:1 was seeded into the sterilized microchambers at a total cell concentration of 3×10^5 cells/mL by gently injecting the cell suspension into the culture chambers with a 1-mL syringe. After incubation overnight at 37 °C and 5% CO₂ for cell attachment, fresh culture media was continuously introduced into the chambers at a flow rate of 0.8 µL/min. Time-resolved progression of cell growth in the microchamber was recorded by a Nikon timelapse inverted microscope in the scanning mode (Yang et al. 2015).

2.1.3 Drug treatment

The clinically approved anti-cancer agent, doxorubicin (DOX) was chosen as a model chemotherapeutic compound. DOX at a concentration of 10 µM in cell culture media was loaded into a 1-mL syringe and continuously introduced into microfluidic chambers through the inlets by a digital pump. We analyzed the cell viability at 0, 4, 8, 12, 16, 20, 24 h after DOX treatment, respectively, using the CellTiter-Blue assay by collecting metabolic wastes from each outlet of the microfluidic chambers and measuring their fluorescence intensity at 560 nm (Ex)/590 nm (Em). Cultures without any treatment were used as controls. While the DOX was introduced to the microchamber at an inlet flow rate of 0.8 µL/min for 24 h, the total amount of DOX applied to the cultured tissue is calculated to be 11.52 µmol.

2.2 Computational model

2.2.1 Computational domain

CompuCell3D (CC3D, v.3.7.5, <https://compucell3d.org/>) was employed for modelling avascular tumor progression and drug response in the microfluidic device. CC3D is an open-source software for simulating the behavior of individual cells and their collective dynamics in a spatial context by the CPM approach (Swat et al. 2012). It also allows users to create custom models by defining cell types and the underlying spatial domain.

To translate experimental data to CC3D simulations, we first converted a representative microscopic image of the microchamber tissue into a 2D lattice format. Briefly, a bright-field image of cells after dynamic culture in a microfluidic chamber on Day 4 was captured and post-processed using ImageJ. We applied an appropriate threshold to generate a binary image of the microscopic image, wherein the cells presented as black entities against a white background. We refined the threshold parameters by comparing the processed image with the original one. Once the cells were segmented, specific size and circularity criteria were set to exclude objects that do not match the cells' morphology. We

also employed an alternative manual segmentation technique through the use of Adobe Photoshop 2020. The clustering configuration of MCF-7 cells presented a discernible contrast, facilitating the use of the 'Magic Wand' tool to select cell colonies based on the inherent contrast. This preliminary selection was then meticulously fine-tuned to match the cell boundaries as observed in the original image. Using the layering function, we were able to effectively segregate and label the different cell types. The post-processed image was imported into CellDraw, a graphical tool used for configuring the initial cell lattice. The drawing was discretized into rectangular blocks and converted into the Potts Initialization File Format (PIFF), where cell regions were coded and cell types were assigned. The unit is in cubic pixel dimensions for volume quantification. In the simulation model of tumor spheroid growth, the lattice dimension was set to $256 \times 256 \times 1$. Following a previous study (Swat et al. 2015), a voxel was set to $4^2 = 16$, and the size of biological cells in the simulation domain represented $16 \mu\text{m}^2$, which generally matches that of real cells. In the microfluidic culture simulation, the prepared PIFF file containing the initialized cell information in a $500 \times 500 \times 1$ lattice domain was then read by CC3D for simulations. The boundaries for the lattice-based simulation were directly defined by the microchamber configuration, with the elongated hexagonal zone designated for cell growth.

2.2.2 The cellular Potts model (CPM)

The CPM complements the lattice with the Glaizer–Graner–Hogeweg (GGH) effective energy to control diverse cell behaviors, such as cell adhesion, cell deformability, and cell–cell interactions (Swat et al. 2012). Briefly, a grid cell from the above PIFF file was chosen randomly and assigned an index $\sigma(i, j)$ as well as a corresponding cell type $\tau(\sigma)$. The simulation proceeded based on a Monte Carlo method by repeating a basic operation, named an index-copy attempt (Jiang et al. 2005; Andasari et al. 2012; Swat et al. 2012). For each attempt, the cell tries to copy its index $\sigma(i, j)$ onto a neighboring grid cell, $\sigma_{neighbor}(i, j)$. As cells tend to evolve towards configurations that minimize GGH effective energy, the index-copy process is constrained following the Hamiltonian function (Eq. 1):

$$H = \sum_{i,j \text{ neighbors}} J(\tau(\sigma(i)), \tau(\sigma(j)))(1 - \delta(\sigma(i), \sigma(j))) + \sum_{\sigma} \lambda_{vol}(\sigma)(v(\sigma) - V_i(\sigma))^2 \quad (1)$$

which involves a boundary energy term and a volume constraint term (Szabó and Merks 2013; Li and Lowengrub 2014; Jafari Nivlouei et al. 2022). The Monte Carlo

simulation accepts an index-copy attempt with probability given by the Boltzmann acceptance function (Eq. 2):

$$P = \begin{cases} 1, \Delta H \leq 0 \\ e^{-\Delta H/T_m}, \Delta H > 0 \end{cases} \quad (2)$$

where ΔH is the change in the effective energy. If $\Delta H \leq 0$, then the attempt will be successful; if $\Delta H > 0$, the attempt will be successful with the probability $e^{-\Delta H/T_m}$. T_m defines the amplitude of membrane fluctuation, which is relative to the constraint strength value of cell property (λ_{vol}).

The Monte Carlo Step (MCS) is the time measure in the simulations. MCS represents a simulated ‘time step’ within the model, comprising attempted cellular interactions and movements. The correlation between MCS and actual physical time is a function of the specific parameters and rules defined within the biologically relevant simulations, such as cell motility rates, division times, or interaction strengths, and the conversion factor is model-dependent. (Swat et al. 2012; Szabó and Merks 2013; Li and Lowengrub 2014; Truong et al. 2016; Sego et al. 2020; Jafari Nivlouei et al. 2022).

2.2.3 Flow field distribution

In addition to the cell characteristic constraints for tumor growth, environmental constraints can also affect tumor behaviors. For example, during the avascular phase, nutrients and chemotherapeutic drugs transport into tumor tissues via diffusion, and their spatial concentrations can be determined using CFD. The steady-state flow field in the microchamber was modeled by the Navier–Stokes equation:

$$\rho \left[\frac{\partial V}{\partial t} + (V \cdot \nabla)V \right] = -\nabla P + \mu \nabla^2 V + \rho g \quad (3)$$

where ρ is the density, V is the velocity vector, μ is the viscosity, P is the pressure, and g is the gravitational acceleration (Hu and Li 2007; Taghibakhshi et al. 2019; Yang et al. 2020).

Given the microchannel’s small scale, the following simplifying assumptions were applied: (1) the fluid flow is completely determined by the pressure distribution and the incompressibility constraint equations. (2) The boundary conditions are no-slip. (3) The flow is quasi-static. Therefore, the time derivative $\rho \left(\frac{\partial V}{\partial t} \right)$ in Eq. (3) can be neglected. Since the hydrostatic pressure difference across the height of the microchamber (200 μm) is minimal, the gravity effect on the flow field (ρg in Eq. 3) can also be considered negligible. Moreover, the flow in microfluidic channels usually operates under a low Reynolds number (Re):

$$Re = \frac{\rho |V| d}{\mu} \approx \frac{|\rho(V \cdot \nabla)V|}{|\mu \nabla^2 V|} \ll 1 \quad (4)$$

Low Reynolds number flows are primarily dominated by viscous forces, hence the term $\rho(V \cdot \nabla)V$ in Eq. (3) can also be ignored.

Taken together, the Navier–Stokes equation in Eq. (3) can be reduced to the Poiseuille flow equation for a Newtonian fluid:

$$\mu \nabla^2 V = \nabla P \quad (5)$$

These simplifications lead to more straightforward computation or analytical solutions that effectively describe the fluid dynamics in the microfluidic device. In this study, we examined the avascular tumor progression by investigating the impact of glucose, the main growth nutrient in cell culture media. The tumor microenvironment was described as a continuous entity, and the glucose distribution in the microfluidic channel was first simulated with the use of continuity equation (Eq. 6) (Mascheroni et al. 2018; Taghibakhshi et al. 2019; Jafari Nivlouei et al. 2022), coupled with the Navier–Stokes equations (Eq. 3).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (6)$$

Time variation of glucose concentration at different lattice points with the presence of cells was modelled by the reaction–diffusion partial differential equations (PDEs) (Jafari Nivlouei et al. 2022):

$$\frac{\partial C_{GLU}(x)}{\partial t} = D_G \nabla^2 C_{GLU}(x) + U_{GLU}(x) + S_G \quad (7)$$

where $C_{GLU}(x)$ is the local glucose concentration, and D_G represents the diffusion coefficient for glucose. The term $U_{GLU}(x)$ denotes the uptake of glucose by cells. S_G represents the secretion coefficient of glucose from the matrix, indicating the release rate of glucose.

2.2.4 Nutrient-dependent tumor growth

The term $U_{GLU}(x)$ in Eq. (7) describes the rate at which cells consume glucose at their lattice point. This rate is determined by a Michaelis–Menten function:

$$U_{GLU}(x) = -\frac{u_{\max}(\tau, s) C_{GLU}(x)}{C_{GLU}(x) + K(\tau, s)} \quad (8)$$

where $u_{\max}(\tau, s)$ is the maximum glucose uptake rate, and $K(\tau, s)$ is the Michaelis constant, both are dependent upon the cell-type (τ) and/or the cell state (s).

In the glucose-dependent tumor spheroid growth simulation, we have defined three cellular states (s) in the model: proliferative cells (‘Proliferative’), quiescent cells (‘Quiescent’), and necrotic cells (‘Necrotic’). When subpopulations of tumor cells have sufficient levels of nutrients, particularly

glucose in our case, they display a proliferative behavior. A growing cell undergoes mitosis when it reaches its doubling volume, and only cells in the proliferative state are capable of this process. These cells divide through mitosis once they reach twice their predefined target volume (V_{target}). The generalized proliferative cells (*'Proliferative'*) follow a specific growth rate according to the empirical Monod equation:

$$\frac{dV_{target}}{dt} = G_{max} \frac{C_{GLU}(x)}{C_{GLU}(x) + C_{GLU50}} \quad (9)$$

where $\frac{dV_{target}}{dt}$ is the growth rate of the cells for reaching the target volume V_{target} , G_{max} is the maximum cell growth rate, $C_{GLU}(x)$ is the glucose concentration at the centroid of the generalized proliferative cell x at that time, and C_{GLU50} is the Monod half-saturation constant, i.e., the glucose concentration at which the cell growth rate is half of its maximum (Sheraton et al. 2020). In contrast, the generalized quiescent cells (*'Quiescent'*) and generalized necrotic cells (*'Necrotic'*) do not undergo changes in target volume.

In the tumor spheroid growth simulation model, the transformation of cellular state is solely dependent on the concentration of glucose. Proliferative cells only grow when the glucose concentration at their centroid is above a certain threshold. Under low glucose conditions, cells that are deprived of glucose gradually accumulate damage, and the rate of damage accumulation depends on the local nutrient concentration. We use the damage accumulation factor to represent the cumulative impact of low nutrient levels on the cells. In each MCS, the glucose concentration at the centroid of each cell is measured. If the concentration falls below the cell's transition glucose concentration threshold, the cell starts to accumulate damage and the damage factor increments. Relevant literature indicates that cellular damage would accumulate when the glucose concentration falls below 0.06 mM (Lieberthal et al. 1998; Pourhasanzade and Sabzpoushan 2021), leading us to set this concentration as the threshold for the cellular state transition from *'Proliferative'* to *'Quiescent'* (P_Q threshold). It is assumed that quiescent cells have a greater resistance to starvation compared to proliferative cells, with their state transition threshold from *'Quiescent'* to *'Necrotic'* (P_Q threshold) set at 0.03 mM. When the cellular damage factor exceeds a certain threshold, an irreversible state transition occurs. Proliferating cells (*'Proliferative'*) shift to a quiescent state (*'Quiescent'*), and quiescent cells transition to a necrotic state (*'Necrotic'*). After mitosis, the accumulated damage in the parent cell is reset to zero, and the daughter cells do not inherit any damage from the parent cell.

The increment of cellular damage accumulation, $\Delta H(x)$, is defined by the following formula:

$$\Delta H(x) = \begin{cases} \frac{C_{GLU}^{threshold} - C_{GLU}(x)}{C_{GLU}^{threshold}}, & C_{GLU}(x) \leq C_{GLU}^{threshold} \\ 0, & C_{GLU}(x) > C_{GLU}^{threshold} \end{cases} \quad (10)$$

Equation (10) includes $C_{GLU}^{threshold}$, which is the glucose concentration threshold for state transition, and $C_{GLU}(x)$ is the glucose concentration at the centroid of the generalized cell x . Subsequently, each generalized cell in the model is iterated upon to ascertain whether the damage factor exceeds the corresponding conversion threshold for that particular generalized cell state. Following the transition, the accumulated damage factor is reset to zero (Table 1).

2.2.5 Drug response

As the time-varying and spatial distributions of drug concentrations can be readily simulated, we further modeled the response of tumor cells to the chemotherapeutic drug DOX in the microfluidic device. The cytotoxic effect of DOX on tumor cells was determined by the spatiotemporal profiles of drug concentration and tumor cell activity. We previously obtained the pharmacokinetics (PK) and pharmacodynamics (PD) data of DOX in the microfluidic system and developed a regression model that describes the correlation between DOX concentration and cell viability. In the present simulation, we defined five different cell types: $\tau = 0$ (medium), $\tau = 1$ (proliferating cell), $\tau = 2$ (quiescent cell), $\tau = 3$ (necrotic cell), and $\tau = 4$ (microfluidic device, freeze). While DOX treatment most likely induced cell cycle transition from proliferating to quiescent or necrotic, cells exposed to DOX also had the chance to repair themselves or develop resistance. Cells were assumed to stop proliferating or undergo apoptosis/necrosis with a probability P , which was directly adapted from the coefficients in the regression model from experimental results. Other parameters used in the simulations were adjusted based on previous literature or through a trial-and-error process with multiple simulation runs to obtain the optimal parameter values that best represent the experimental findings.

2.2.6 Computational methodology

We utilized CC3D to simulate the progression of avascular tumors and their response to drugs in a microfluidic device. CC3D is a computational framework based on the GGH model, which is integrated with a PDE solver, written in Python. The input parameters, including the biophysics of cells and the properties of extracellular materials, were used to define the behavior and dynamics of cells such as growth, proliferation, and mitosis. The CC3DML file within the software was used to specify the energy terms, diffusion solvers, and initial cell layout, respectively.

Table 1 Simulation terms and parameter estimations for the avascular tumor spheroid growth model

Term	Description	Value
τ, s	Cell type, Cell state	0 = Medium 1 = MCF-7, Proliferative 2 = MCF-7, Quiescent 3 = MCF-7, Necrotic
Lattice dimension		$256 \times 256 \times 1$
T_m	Cell membrane fluctuation	20
λ	Constraint strength value	10
$V_I(\sigma(\tau))$	Initial volume	16
$V_T(\sigma(\tau))$	Cell doubling volume, $\tau = 1$	32
S_G	Secretion coefficient of glucose from the matrix	0.2, 0.175, 0.15, 0.125
D_G	Diffusion constant of glucose	4500 voxel/MCS
P_Q threshold	Glucose concentration threshold for P_Q transition	0.06 mM
P_Q damage factor threshold	Accumulation damage threshold for P_Q transition	256
Q_N threshold	Glucose concentration threshold for Q_N transition	0.03 mM
Q_N damage factor threshold	Accumulation damage threshold for Q_N transition	512
$u_{\max}(\tau)$	Maximum glucose uptake rate	0.75, $\tau = 1$ 0.5625, $\tau = 2$ 0, $\tau = 3$
$K(\tau)$	Michaelis constant	0.00256, $\tau = 1$ 0.00256, $\tau = 2$ 0, $\tau = 3$

3 Results and discussion

3.1 Microfluidic fabrication and cell growth

Figure 1a,b depicts the configuration of the fabricated microfluidic device comprising eight elongated hexagonal prism-shaped chambers connected to two microchannels as the inlet and outlet, respectively. The height of the microfluidic chamber is 200 μm , and the microchamber has total volume of 10 μL . Breast cancer cells (MCF-7) were co-cultured with adipose stromal cells (ASC) to better emulate the in vivo physiologic conditions, where tumor cells were surrounded by adipocytes. Time-resolved progression of cell growth in the microfluidic device was monitored and recorded by a microscope as shown in Fig. 1c, illustrating the sequential steps of tumor cell clustering. After cell seeding, over 50–60% of cells became confluency over the microchamber surface on Day 4. Starting from Day 7, noticeable multicellular acini-like aggregates began to form, while nodule-like structures were observed by Day 9. The formation of cell clusters and the development of nodule-like structures over time are marked with # and the arrows indicate the edges of the formed nodules in the enlarged areas in Fig. 1c. The entire microchamber was fully occupied by the tumor tissues after a 2-week-long dynamic culture. This 3D architecture was further confirmed through hematoxylin and eosin (H&E) staining, where the cross-sectional staining of the harvested tissue

from the microchamber revealed a compact arrangement of cells, highlighting the cellular 3D organization (data not shown) (Yang et al. 2015).

Long-term cell growth in the microchambers was supported by continuous introduction of fresh cell culture media using the syringe pumps with controllable flow rates. In this study, an inlet flow rate of 0.8 $\mu\text{L}/\text{min}$ was selected based on our previous CFD simulation and experimental results. The calculated average shear stress at the central region of the microfluidic chamber is approximately 1.92×10^{-4} $\text{dyne}\cdot\text{cm}^{-2}$, with an integral shear stress reaching 6.56×10^{-2} $\text{dyne}\cdot\text{cm}^{-2}$, which falls within the physiological range (10^{-6} to 0.1 $\text{dyne}\cdot\text{cm}^{-2}$) induced by the interstitial flows in a compact avascular tumor tissue (Mitchell and King 2013a,b). Due to the high cellular density and associated extracellular matrix, fluid shear stress values within the tumor interstitium are significantly lower than the magnitudes within the circulatory system. Thus, we posit that the primary mechanism responsible for supporting the long-term cell growth in the microfluidic device is the continuous introduction of fresh nutrients. From a technical perspective, during initial cell seeding and dynamic culture in microfluidic systems, cell adhesion to the chamber surface is relatively weak, and we did find higher flow rates increase the risk of cell detachment. We determined that an inlet flow rate of 0.8 $\mu\text{L}/\text{min}$ adequately supported nutrient transport, while simultaneously ensuring cellular viability and maintaining cell integrity within the microfluidic chamber.

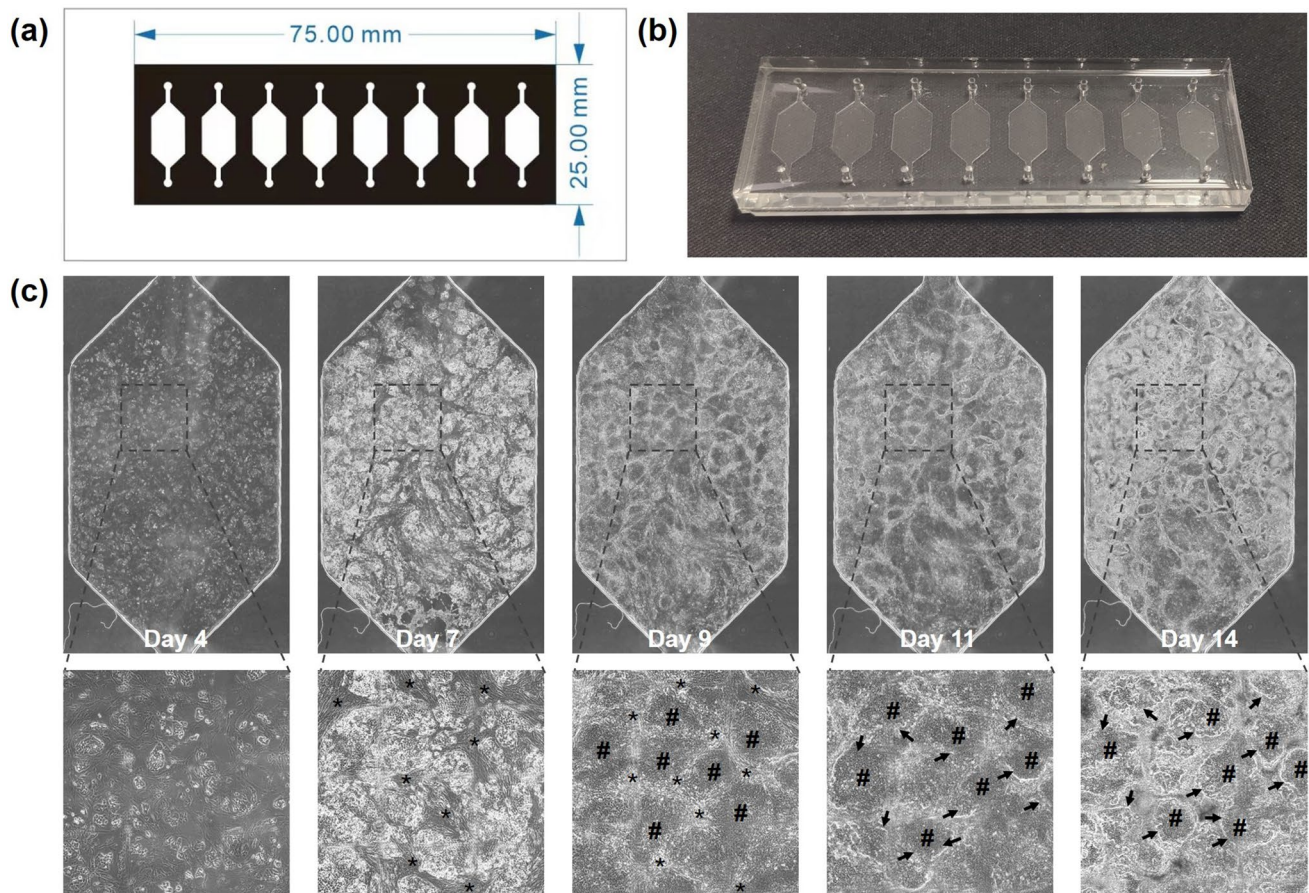


Fig. 1 **a** Design of the schematic layout of the 8-chamber microfluidic device. **b** A top-down view of the fabricated microfluidic device. **c** Time-resolved progression of MCF-7 (#) and ASCs (*) cultured in a microfluidic chamber. Images were taken with a timelapse inverted

microscope in the scanning mode on days 4, 7, 9, and 14. Cell clusters formed and developed into nodule-like structures over time. The arrows indicate the edges of the formed nodules

3.2 Glucose-dependent tumor spheroid growth simulation

During the avascular phase of tumor growth, nutrient substances, such as glucose, are transported in and out of cells through diffusion. The nutrient distribution and kinetics of diffusion are major factors that facilitate cell growth and influence tumor cell behaviors. We firstly simulated the glucose diffusion process along with progression of tumor spheroid growth in both spatial and temporal domains. Figure 2a shows the time-dependent progression of glucose diffusion along with tumor growth. Figure 2b presents a plot of glucose concentration along the radial axis, indicating a higher concentration at the periphery of the 3D tumor spheroids which gradually decreases from the edge to the core. The dense packing of cells in 3D cultures can result in a diminished diffusion rate of nutrients into the spheroid's core. Additionally, we investigated the effects of nutrient availability (i.e., glucose concentration) on

cellular proliferation in a quantitative manner. The simulations were conducted to compare the necrotic regions within the tumor spheroids, as depicted in Fig. 2c. The surrounding glucose supply is represented by the parameter S_G in Eq. (7). It is observed that the necrotic regions within the spheroids progressively increase as the available nutrient decreases (Fig. 2d), suggesting that sufficient medium is critical for cells to retain their viability and proliferation.

To relate the simulation's MCS time-scale to minutes, previous study matched cell-migration speeds in simulations to that obtained in experiments (Swat et al. 2015). For example, given the experimental value of cell migration speed is about $0.4 \mu\text{m}/\text{min}$ and the side of biological cells is $4 \mu\text{m}$, cells in the simulation move at an average speed of $0.1 \text{ pixel}/\text{MCS}$, and one MCS represents 1 min (Swat et al. 2012). In our study, the cell division cycle is assumed to be 24 h, and each MCS in Fig. 2 represents 2 min.

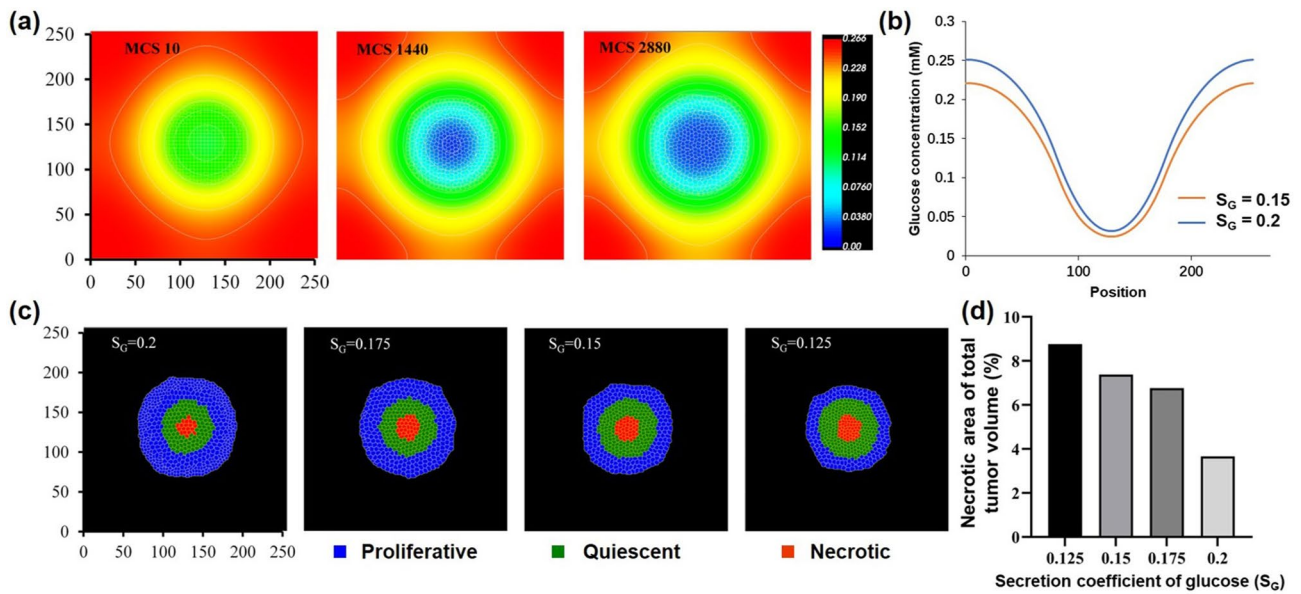


Fig. 2 Glucose-concentration dependent tumor spheroid growth simulation. **a** The distribution of glucose concentration in the tumor spheroid via diffusion. **b** Glucose concentration along the radial axis of the tumor spheroid, indicating higher concentration at the periph-

ery which gradually decreases from the edge to the core. **c** Tumor growth under different nutrient conditions, as represented by the glucose supply parameter S_G . **d** The impact of different glucose conditions on the percentage of necrotic area in the tumor spheroid

3.3 Glucose distribution simulation in the microfluidic chamber

The starting conditions to simulate the glucose distribution in the microchamber and the tumor cell growth were translated directly from the experimental data acquired from the microfluidic cultures in real applications. As shown in Fig. 3a, MCF-7 cells were labeled with a cell-tracking dye so they can be visually identified based on the green fluorescence. This allows for clear identification of the MCF-7 cell colonies, which are morphologically distinguishable from the adjacent ASCs. These morphological distinctions became increasingly evident under microscopic examination as the cells proliferated within the microfluidic environment in Fig. 3b. MCF-7 cells ordinarily are characterized by their polygonal or cobblestone appearance with a much larger circularity index. Conversely, ASCs adopt a fibroblastic phenotype, featuring an elongated, spindle-like shape. In terms of size, MCF-7 cells are generally smaller and display a more consistent diameter relative to stromal cells. Thereafter, the two cell populations, characterized by their disparate morphological attributes, can be distinguished and marked using image analysis software. Details are provided in “Materials and methods” section. Figure 3c showcases the general procedure for converting the microscopic image to the Potts initialization file format (PIFF). The enlarged area of the representative microscopic image schematically shows the segmentation of cell area into a set of grids, where MCF-7 cell clusters (blue

regions) were differentiated from surrounding ASC. A 2D lattice domain was coded with cell positions and defined with respective cell types in each line.

We then simulated the dynamic glucose diffusion process within microchambers in both spatial and temporal domains according to the microchannel design. The boundaries of cellular automata lattices and prohibited zones for simulation were directly derived from the microchamber configuration. The glucose solution was assumed to be introduced at the inlet (top) and diffused through the elongated hexagonal computational domain to the outlet (bottom). Figure 4a shows the contours of concentration fields of glucose in the microfluidic chamber at selected iterations without the presence of cells. A spatial linear distribution of glucose parallel to the flow direction was generated along the central axis of the microchamber through simple free diffusion. A stable and uniform glucose concentration distribution was fully established as the medium perfused through the entire chamber area after reaching equilibrium with time.

Since the physical characteristics of cells primarily influence the rate of diffusion, we then included the reaction–diffusion model to analyze the spatial and temporal distribution of glucose with the presence of cells. In contrast to a simple diffusion scenario shown in Fig. 4a, active uptake and consumption of glucose by cells significantly influenced the glucose diffusion process. The non-linear distribution in Fig. 4b indicates that nutrients may encounter resistance as they enter cells, which in turn delay the diffusion process and affect the uniformity of the nutrient fields.

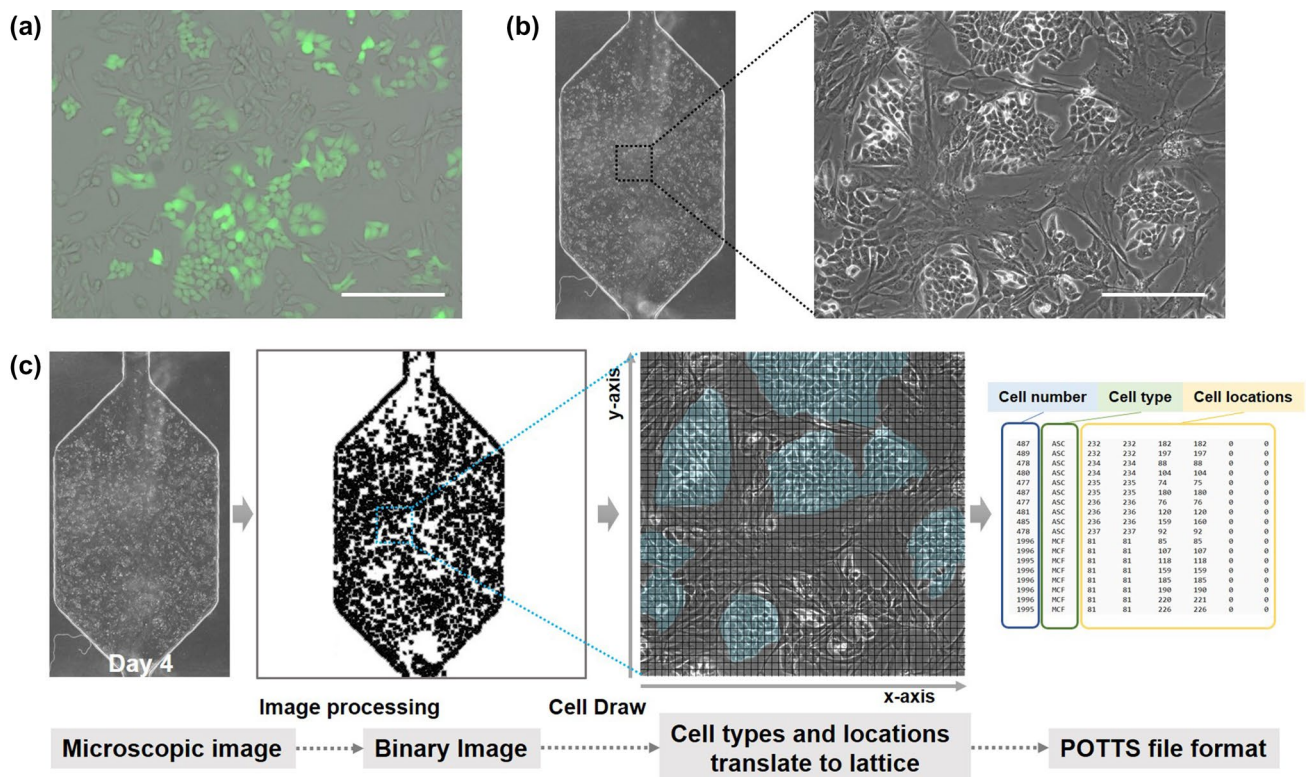


Fig. 3 **a** MCF-7 cells were pre-stained with green fluorescence and co-cultured with ASCs. **b** Enlarged area selected from a representative bright-field image captured from the co-cultures in the microfluidic chamber on Day 4, showing distinguishable cell morphology of different cells. Scale bar 200 μm . **c** Conversion of the microscopic image of the microchamber culture to the lattice domain. The elon-

gated hexagonal microchamber configuration defines the boundaries of the lattice-based simulation, and the placement of cellular pots follows their initial seeding location. A representative area of the microscopic image with magnification schematically shows the gridding of the lattice, where MCF-7 cell aggregates were marked as blue regions

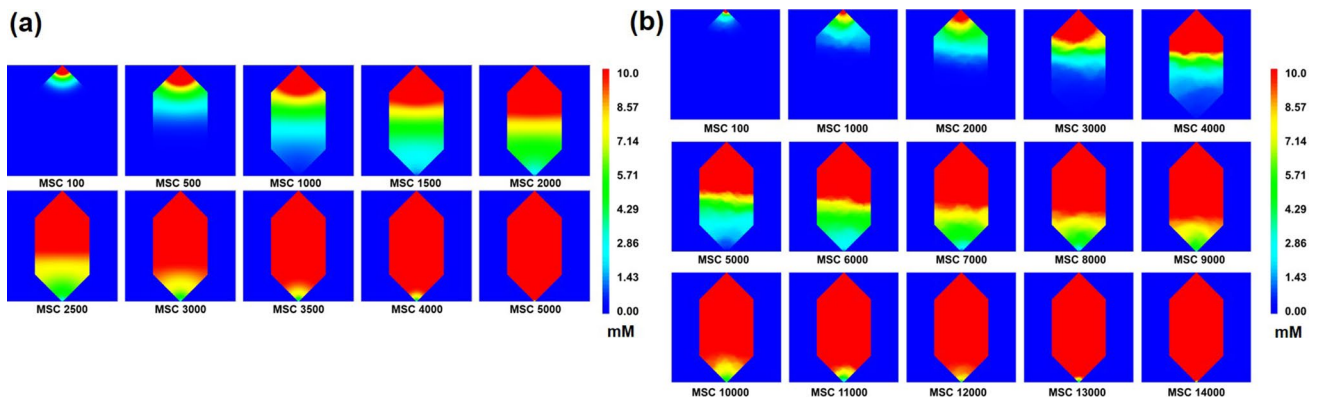


Fig. 4 Pseudo color-coded glucose distribution in the microchamber **a** without the presence of cells, and **b** with the influence of cells. Color bar indicates the range of concentrations of glucose (high values = red, low values = blue)

While the syringe pump did induce a convective flow from the inlet of the microfluidic chamber, in our setup, the inlet flow was primarily used to maintain a constant replenishment of medium nutrients rather than generating significant flow velocities or shear stress. The operating inlet flow rates were chosen such that the hydrodynamic conditions

allow for a quasi-steady-state diffusion scenario, thereby minimizing convective effects within the region. Therefore, our simulation solely considered the influence of diffusion when accounting for the presence of cells in the microchamber. In this case, we were better able to mimic *in vivo* physiological conditions and mirror the experimental findings.

Additionally, this simplification helps us understand the cellular response without further complexity that convective flow might introduce.

3.4 Nutrient-based avascular tumor growth

Figure 5 illustrates the corresponding profiles of cell proliferation along with the simulated tumor growth based on collective dynamic behaviors of cells in the microfluidic chamber. The gray spots in Fig. 5a, b indicate proliferating MCF-7 cells while the purple spots refer to ASCs. Similar to experimental observations (Fig. 1), tumor cells tended to form multiple clusters after seeding in the microchamber, and the tumor cells in those nucleation sites were able to grow in an aggregated manner, thereby increasing the likelihood of forming small spheroid-like structures. With the progression of simulation, the initial tumor clusters continued to grow, migrate and bridge a short distance to remodel and reshape the spheroid structure over time. As the cell growth was confined to the microchamber dimension, the final size of the aggregates reached a maximum volume determined by the size and configuration of the microchamber (Fig. 5c). To establish an approximate correlation between MCS and real-world time, we calculated each MCS in Fig. 5 to represent 36 min, as the available experimental data for simulation covered the period from Day 4 to Day 14 of the microfluidic culture. Notably, this time scale is very large because the initial lattice in the model was taken from microscopy images of the culture after 4 days of growth, and the simulation began with large cell clusters rather than individual cells. The limited resolution can thus significantly affect the time scale of the simulation.

We also observed that the tumor growth at the avascular stage was closely dependent on the concentration of nutrients delivered to the cells. Proliferating cells would continue dividing until they reached a certain volume-to-surface ratio under a sufficient supply of nutrients, and the total volume increased as the tumor cells consumed surrounding glucose. Yet, when the concentration of nutrients dropped below a certain threshold, starving tumor cells became damaged and underwent apoptosis or necrosis. From our simulation results, we indeed observed the nutrient-dependent formation of a necrotic core in the central region of the tumor spheroid (Fig. 2c), which was often found experimentally in traditional static culture conditions (Tang et al. 2014; Mukomoto et al. 2020). We investigated the effects of glucose availability on cellular proliferation in the microfluidic culture by setting the release rate of glucose into the surrounding environment (S_G). Similar to above glucose-dependent tumor spheroid growth simulation, tumor cells also tended to change their status from proliferative (gray) to necrotic (pink) depending on nutrient availability, and necrotic areas gradually developed and expanded as glucose depleted (Fig. 5b). This finding confirmed that proliferating tumor cells in the microfluidic culture required a sufficient nutrient supply to sustain their long-term growth.

Compared to previous computational models, the CPM-based simulations could closely reproduce the observed growth of avascular tumors in the microfluidic cultures by mirroring cell–cell interactions such as adhesion and junctional connections. Additionally, as the cells were fully integrated into an intact and dense tissue layer along with tumor progression, quantifying the total cell number in the microfluidic device during dynamic culture became experimentally challenging. However, computational modeling

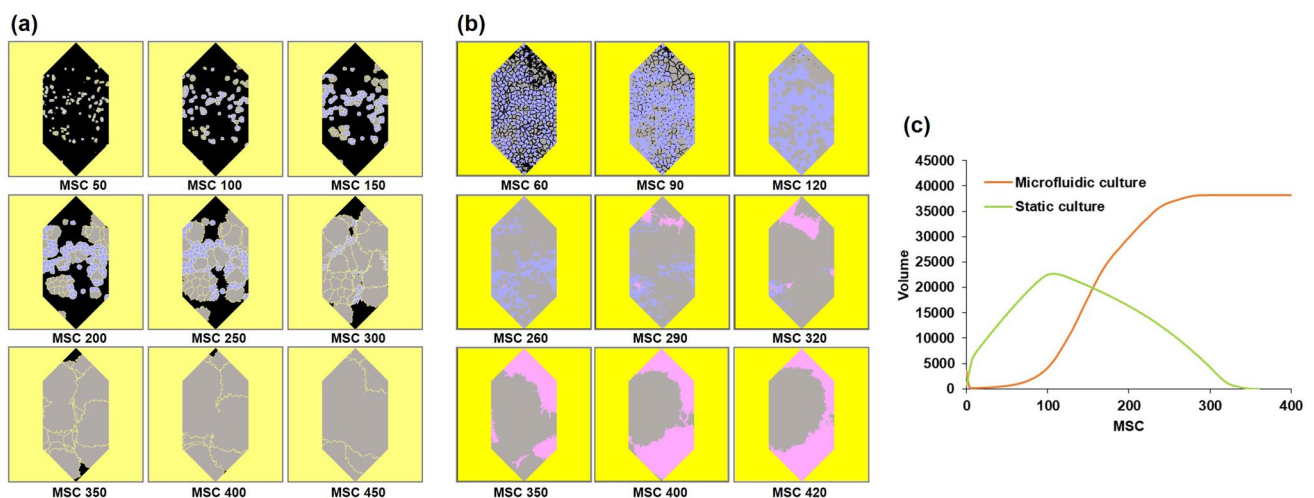


Fig. 5 **a** CPM-based computational simulation results of cell proliferation and shape evolution in the microchamber with continuous medium supply. **b** Simulated avascular tumor growth in the microchamber under limited-glucose conditions. Snapshots of selected rep-

resentative iterations are shown. **c** The changing volume of total cells showing varying growth curves of cells during static and dynamic culture in the microchamber

offered detailed and quantitative information on the cell growth dynamics. Figure 5c illustrates the simulated tumor volume over time, which is very difficult to be measured in the real experiments. To our interests, the simulated proliferation rate of tumor cells in the microfluidic culture followed the Gompertz-like curves, which aligned well with the in vivo tumor growth at the avascular stage. Initially, tumor cells proliferated rapidly, followed by an exponential growth and a subsequent slowing down until reaching the maximum tumor volume as the simulation progressed. The final tumor size was restricted by the confined culture area determined by the microchamber configuration, indicating the possibility of producing uniform yet size-controlled 3D multicellular tumor spheroids within the microfluidic devices, suitable for standardized drug testing. It should be noted that, throughout the duration of the microfluidic culture, the glucose concentration was excessive for the cells as fresh media were continuously introduced, providing an adequate amount of nutrients to maintain normal metabolic activities. However, under nutrient-limited conditions, such as in the static culture without continuous medium supply, the exponential tumor growth phase would be followed by a declining trend. As depicted by the predicted curve in Fig. 5c, once reaching its maximum volume, the tumor tissue would gradually shrink and eventually diminish in size.

3.5 Therapeutic response to DOX treatment

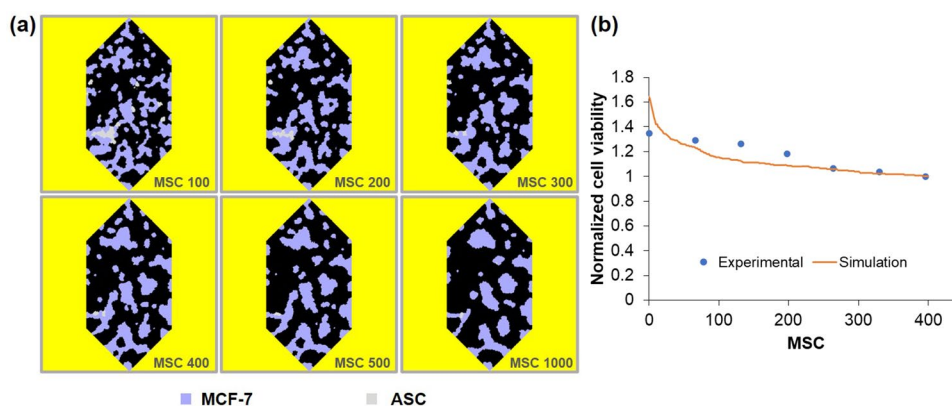
DOX is a clinically approved chemotherapy drug that prevents the growth of cancer cells by either interfering with DNA or generating cytotoxic free radicals. According to previous literatures (Liston and Davis 2017), the peak plasma concentrations of DOX can reach 1–15 μM following a single bolus injection during clinical therapeutics. In our earlier work, we have tested the in vitro therapeutic effectiveness of DOX at various concentrations of 1, 5, and 10 μM in 3D tumor models using microfluidic technology (Yang et al. 2020). The microfluidic 3D tissue model exhibited stronger drug resistance as the dose of DOX increased, in

comparison to traditional 2D static cultures. In the present study, the cytotoxic effects of DOX on the microfluidic cultures were numerically modeled using the CPM approach. The outcomes of drug actions on a local cell population were dependent on the interactions among delivered drug dosage, cell mitosis phase, and cell repair probability (Sheraton et al. 2020). We chose the DOX concentration of 10 μM as it not only owns clinical relevance but also presented the most significant differences in cell viability between the culture models.

We directly implemented the correlation between DOX dosage and cell viability obtained from experiments in the simulation model. As shown in Fig. 6a, the simulated cell proliferation and changes in tumor morphology in the microchamber were captured at various time intervals. In the simulation domain, DOX and cell culture medium were simultaneously introduced into the microchannels. As cells would continue multiply under the conditions with sufficient growth nutrients, it was not surprising to see the retention of proliferating cells (purple color). However, the coverage of proliferating cells in the microchambers decreased upon continuous exposure to the drug, suggesting the therapeutic efficacy of DOX in inhibiting cell proliferation. Furthermore, according to the simulated mass diffusion profile, it was expected that DOX would distribute uniformly throughout the microchamber. For each nodule-like structure, DOX directly acted upon cells located in the peripheral regions, thereafter, leading to a gradual shrinkage of corresponding nodule areas during the course of treatment. Our previous fluorescence images of the Live/Dead staining of the microfluidic cell cultures after DOX treatment also showed apparent cell death in the outer regions of the tumor nodules, while more viable cells remained in the core area (Yang et al. 2020). This could be attributed to the higher likelihood of DOX exposure to the cells located in the superficial region in comparison to those in the core.

The plot of tumor volume for varying time steps is depicted in Fig. 6b and compared with our previous experimental observations in which the suppression of cell growth

Fig. 6 **a** Contour plots of proliferating cells in the microchambers under DOX treatment over time. **b** Comparison of experimental data and simulation results for normalized cell viability after DOX treatment



after DOX treatment was quantified at 0, 4, 8, 12, 16, 20, 24 h, respectively. The simulated values of normalized total tumor volume were found to correspond well to the experimental values in terms of cell viability in the DOX-treated microfluidic culture at all time points. DOX is known to bind to DNA and inhibit the functions of topoisomerase II, an enzyme necessary for DNA replication and repair (Nitiss 2009). Thus, DOX treatment would arrest cell cycle at the G2/M phase and prevent cells from dividing and growing. As shown in Fig. 6b, tumor cell proliferation could be significantly inhibited during the acute DOX exposure period, yet the declining trend slows down with time. Similar to what was observed experimentally, 3D tumor tissue grown in the microchambers may develop resistance to the chemotherapeutic drugs along with their self-repairing mechanism, leading to a lower sensitivity to DOX during the course of treatment.

3.6 Limitation of the current study

The simulated results in our study, based on CAM using CC3D, coincided well with experimental measurements, and would provide potential hints on cell variability in response to different treatments. However, the model in this study was oversimplified by only accounting the basic cell functions of cells and this should be further adjusted in the future. We used a 2D cell-lattice domain in simulations, mainly considering the computational efficiency and the relevance to experimental data. When 3D architecture and cellular dynamics were achievable experimentally, a 3D model will likely be more appropriate despite the increased demand for computational power. While the present work was confined to the MCF-7-derived microfluidic tumor tissue models, extending our framework through the inclusion of the results from other cancer cell lines will fortify the generalizability of our findings. Regarding the conversion of voxel and MCS in the simulation domain into the real-world scale, the spatiotemporal resolution can be improved by incorporating more information from the experimental side, such as visualization of single cell movement in the microchamber. The cell growth rates and diffusion coefficient of key incorporated substances (i.e., glucose and DOX) used in the simulations also need to be further optimized according to the real experimental data. Furthermore, the missing parameters such as the influence of oxygen diffusion should be included to further improve the prediction accuracy of the simulations. Although tumor cells at the avascular stage are not typically subject to strong directional flows and associated shear stresses, they may be exposed to shear forces exerted on the tumor cell surfaces adjacent to blood vessels or lymphatic flows. In these scenarios, apart from the diffusion-only mass transfer, the simulation should also integrate the effects of continuous-flow perfusion and

convection-mediated mass transport exerting on tumor spheroids. For example, data from FEM-based flow distribution could be combined with the cellular automata lattice as input, allowing for dynamic adjustment of nutrient/drug profiles within the microchamber.

4 Conclusion

The computational simulation presented in this study can stand out both through its experimental origins and its ability to predict the tumor progression as well as therapeutic outcomes within a microfluidic apparatus. The numerical model was formulated based on experimental evidence pertaining to the configuration of microchambers, microscopic observations of cellular morphology and spatial distribution, as well as the temporal alterations in cellular proliferation rates upon drug actions. Employing the CPM approach, this study predicted the spatiotemporal evolution of tumor progression in the microfluidic device with quantitative information on cell growth dynamics and reciprocal interactions with surrounding environment. The simulated results were validated against experimental data, paving the way towards in-silico determination of optimal drug dosages during therapeutic screening and testing.

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Author contributions S.L.: data curation, methodology, writing—original draft. Y. L.: data curation, software, validation. C.C.: supervision, project administration. Z.Q.: supervision, funding acquisition. H.W.: writing—review and editing. Y.Y.: conceptualization, writing—review and editing.

Data availability The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

References

- Andasari V, Roper RT, Swat MH, Chaplain MAJ (2012) Integrating intracellular dynamics using CompuCell 3D and bionetsolver: applications to multiscale modelling of cancer cell growth and invasion. *PLoS One* 7:e33726. <https://doi.org/10.1371/journal.pone.0033726>
- Ballesteros Hernando J, Ramos Gómez M, Díaz Lantada A (2019) Modeling living cells within microfluidic systems using cellular automata models. *Sci Rep* 9:14886. <https://doi.org/10.1038/s41598-019-51494-1>

- Carvalho V, Rodrigues RO, Lima RA, Teixeira S (2021) Computational simulations in advanced microfluidic devices: a review. *Micromachines* 12:1149. <https://doi.org/10.3390/mi12101149>
- Chen Y, Wang H, Zhang J et al (2015) Simulation of avascular tumor growth by agent-based game model involving phenotype-phenotype interactions. *Sci Rep* 5:17992. <https://doi.org/10.1038/srep17992>
- Dong R, Li Y, Liu S et al (2022) On-chip spectroscopic monitoring of erythrocyte oxygenation under hematocrit and oxygen gradients. *J Sci Adv Mater Devices* 7:100515. <https://doi.org/10.1016/j.jsamd.2022.100515>
- Graner F, Glazier JA (1992) Simulation of biological cell sorting using a two-dimensional extended Potts model. *Phys Rev Lett* 69:2013–2016. <https://doi.org/10.1103/PhysRevLett.69.2013>
- Han SJ, Kwon S, Kim KS (2021) Challenges of applying multicellular tumor spheroids in preclinical phase. *Cancer Cell Int* 21:152. <https://doi.org/10.1186/s12935-021-01853-8>
- Hu G, Li D (2007) Three-dimensional modeling of transport of nutrients for multicellular tumor spheroid culture in a microchannel. *Biomed Microdevices* 9:315–323. <https://doi.org/10.1007/s10544-006-9035-1>
- Jafari Nivlouei S, Soltani M, Shirani E et al (2022) A multiscale cell-based model of tumor growth for chemotherapy assessment and tumor-targeted therapy through a 3D computational approach. *Cell Prolif* 55:e13187. <https://doi.org/10.1111/cpr.13187>
- Jiang Y, Pjesivac-Grbovic J, Cantrell C, Freyer JP (2005) A multi-scale model for avascular tumor growth. *Biophys J* 89:3884–3894. <https://doi.org/10.1529/biophysj.105.060640>
- Kaura P, Mishra T, Verma N et al (2021) Effects of combined chemotherapeutic drugs on the growth and survival of cancerous tumours—an in-silico study. *J Comput Sci* 54:101421. <https://doi.org/10.1016/j.jocs.2021.101421>
- Li JF, Lowengrub J (2014) The effects of cell compressibility, motility and contact inhibition on the growth of tumor cell clusters using the Cellular Potts Model. *J Theor Biol* 343:79–91. <https://doi.org/10.1016/j.jtbi.2013.10.008>
- Lieberthal W, Menza SA, Levine JS (1998) Graded ATP depletion can cause necrosis or apoptosis of cultured mouse proximal tubular cells. *Am J Physiol Ren Physiol*. <https://doi.org/10.1152/ajprenal.1998.274.2.f315>
- Liston DR, Davis M (2017) Clinically relevant concentrations of anti-cancer drugs: a guide for nonclinical studies. *Clin Cancer Res* 23:3489–3498. <https://doi.org/10.1158/1078-0432.CCR-16-3083>
- Mascheroni P, Carfagna M, Grillo A et al (2018) An avascular tumor growth model based on porous media mechanics and evolving natural states. *Math Mech Solids* 23:686–712. <https://doi.org/10.1177/1081286517711217>
- Milotti E, Chignola R (2010) Emergent properties of tumor micro-environment in a real-life model of multicell tumor spheroids. *PLoS One* 5:e13942. <https://doi.org/10.1371/journal.pone.0013942>
- Mitchell MJ, King MR (2013a) Fluid shear stress sensitizes cancer cells to receptor-mediated apoptosis via trimeric death receptors. *New J Phys* 18:015008. <https://doi.org/10.1088/1367-2630/15/1/015008>
- Mitchell MJ, King MR (2013b) Computational and experimental models of cancer cell response to fluid shear stress. *Front Oncol* 3:1–11. <https://doi.org/10.3389/fonc.2013.00044>
- Mukamoto R, Nashimoto Y, Terai T et al (2020) Oxygen consumption rate of tumour spheroids during necrotic-like core formation. *Analyst* 145:6342–6348. <https://doi.org/10.1039/d0an00979b>
- Nitiss JL (2009) Targeting DNA topoisomerase II in cancer chemotherapy. *Nat Rev Cancer* 9:338–350. <https://doi.org/10.1038/nrc2607>
- Pourhasanzade F, Sabzpoushan SH (2021) A new mathematical model for controlling tumor growth based on microenvironment acidity and oxygen concentration. *Biomed Res Int*. <https://doi.org/10.1155/2021/8886050>
- Sadhukhan S, Basu SK (2020) Avascular tumour growth models based on anomalous diffusion. *J Biol Phys* 46:67–94. <https://doi.org/10.1007/s10867-020-09541-w>
- Sego TJ, Aponte-Serrano JO, Gianlupi JF et al (2020) A modular framework for multiscale, multicellular, spatiotemporal modeling of acute primary viral infection and immune response in epithelial tissues and its application to drug therapy timing and effectiveness. *PLoS Comput Biol* 16:e1008451. <https://doi.org/10.1371/journal.pcbi.1008451>
- Sheraton MV, Chiew GGY, Melnikov V et al (2020) Emergence of spatio-temporal variations in chemotherapeutic drug efficacy: in-vitro and in-silico 3D tumour spheroid studies. *BMC Cancer* 20:1201. <https://doi.org/10.1186/s12885-020-07677-5>
- Swat MH, Thomas GL, Belmonte JM et al (2012) Multi-scale modeling of tissues using CompuCell 3D. *Methods Cell Biol* 110:325–366. <https://doi.org/10.1016/B978-0-12-388403-9.00013-8>
- Swat MH, Thomas GL, Shirinifard A et al (2015) Emergent stratification in solid tumors selects for reduced cohesion of tumor cells: a multi-cell, virtual-tissue model of tumor evolution using CompuCell 3D. *PLoS One*. <https://doi.org/10.1371/journal.pone.0127972>
- Szabó A, Merks RMH (2013) Cellular Potts modeling of tumor growth, tumor invasion, and tumor evolution. *Front Oncol* 3:87. <https://doi.org/10.3389/fonc.2013.00087>
- Taghibakhshi A, Barisam M, Saidi MS et al (2019) Three-dimensional modeling of avascular tumor growth in both static and dynamic culture platforms. *Micromachines* 10:580. <https://doi.org/10.3390/mi10090580>
- Tang L, Van De Ven AL, Guo D et al (2014) Computational modeling of 3D tumor growth and angiogenesis for chemotherapy evaluation. *PLoS One* 9:e83962. <https://doi.org/10.1371/journal.pone.0083962>
- Truong D, Puleo J, Llave A et al (2016) Breast cancer cell invasion into a three dimensional tumor-stroma microenvironment. *Sci Rep* 6:34094. <https://doi.org/10.1038/srep34094>
- Uhl CG, Liu Y (2019) Microfluidic device for expedited tumor growth towards drug evaluation. *Lab Chip* 19:1458–1470. <https://doi.org/10.1039/c8lc01250d>
- Yang Y, Yang X, Zou J et al (2015) Evaluation of photodynamic therapy efficiency using an in vitro three-dimensional microfluidic breast cancer tissue model. *Lab Chip* 15:735–744. <https://doi.org/10.1039/c4lc01065e>
- Yang Y, Liu S, Chen C et al (2020) Microfluidic-enabled self-organized tumor model for in vitro cytotoxicity assessment of doxorubicin. *Biomed Microdevices* 22:70. <https://doi.org/10.1007/s10544-020-00523-2>
- Yang Y, Geng J, Zhang H et al (2021) Image-guided simulation in comparison with laser speckle contrast imaging for full-field observation of blood flow in a microvasculature model. *Microvasc Res* 133:104092. <https://doi.org/10.1016/j.mvr.2020.104092>
- Yang Y, Dong R, Zhang S et al (2022) A microfluidic system for viability determination of microalgae upon disinfectant treatment under continuous flow. *Sci Total Environ* 816:151615. <https://doi.org/10.1016/j.scitotenv.2021.151615>

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